Microscopic modeling of stress-induced degradation and dielectric breakdown in MTJ barrier for high-density STT-MRAM

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We present a microscopic model for stress-induced degradation and dielectric breakdown of the MgO barrier in magnetic tunnel junctions (MTJs). By constructing a Markov model based on density functional theory (DFT) calculations, we investigate how different impurity atoms in the barrier affect dielectric breakdown at the atomic scale. We show that impurities with higher electronegativity, such as Ir and Pt, substantially shorten breakdown lifetimes by enhancing oxygen-vacancy formation. Reducing such impurities is critical for improving the reliability of spin-transfer torque MRAM (STT-MRAM). Using the same simulation method, we also find that resistance drift and time-dependent degradation of tunnel magnetoresistance (TMR) ratio can be explained by the generation of oxygen Frenkel defects at the MgO interface. These findings provide an atomistic perspective that refines traditional phenomenological models of stress-induced degradation and dielectric breakdown for MTJs.

***Index Terms*—STT-MRAM, TDDB, DFT calculations, impurity.**

I. Introduction

STT-MRAM has attracted attention as a promising non-volatile memory device with high speed, high endurance, and low power consumption. Recent advances in scaling of MTJs [1], [2] have expanded the potential applications of STT-MRAM including high-density nonvolatile main memory and storage-class memory. A key challenge in scaling MTJs is improving the endurance of the MgO barrier layer. As MTJ diameters shrink, the MgO barrier becomes increasingly susceptible to dielectric breakdown [3] because a thinner MgO layer is required for STT switching at the same operating voltage. Thinning the MgO barrier layer also results in resistance drift and TMR degradation under a voltage stress [4]. The microscopic mechanisms and suppression methods of the resistance drift and TMR degradation have not been established. Moreover, previous studies have reported that the endurance is degraded by impurities diffusing from other layers in the MTJ stack [5], [6]. Nevertheless, the microscopic mechanisms by which these impurities affect dielectric breakdown have not been investigated in detail, and it remains unclear which impurities most critically affect reliability.

We propose a microscopic model for stress-induced degradation and dielectric breakdown of the MgO barrier by using a Markov model based on DFT calculations. We theoretically clarify the impact of various impurity atoms on the dielectric breakdown of MTJs [7]. Also, the mechanisms and suppression methods of the resistance drift and TMR degradation are investigated [8].

II. Simulation Methods and Model

We assume that applying a constant voltage generates oxygen vacancies in the MgO layer, eventually leading to breakdown. To obtain the breakdown lifetime and degradation behavior, we use the simulation model shown in Fig. 1. The formation and growth of a percolation path of oxygen vacancies are represented by transitions through a series of states. The transition rates between states depend on current density, applied voltage, and defect formation energies. The fraction of each state in the entire MTJ evolves over time according to the Markov model. The model parameters, such as the defect formation energies and $I-V$ characteristics of each state, are obtained from DFT calculations and DFT-based non-equilibrium Green function calculations.

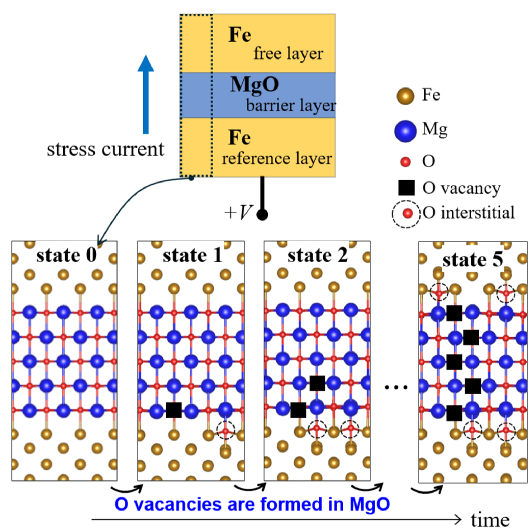

Fig. 1. Proposed breakdown model. Oxygen vacancies and oxygen interstitials are generated under constant voltage stress. In each local region, the state changes step by step from state 0 to state 5. Adapted from [7]. ©2026 IEEE.

III. Results and Discussion

Fig. 2(a) shows the time dependence of the MTJ resistance under a constant voltage stress obtained from simulations with our Markov model. Oxygen vacancy density increases over time under stress, leading to a reduction in resistance in both the parallel (P) and antiparallel (AP) states. The TMR ratio and the AP resistance begin to decrease earlier than the P resistance because conduction in the AP state is more susceptible to conductance changes induced by oxygen vacancies. As vacancies accumulate, O interstitials are also generated and energetically stable at the interface between MgO and Fe. The accumulation of these interstitials produces an increase in resistance prior to breakdown. We find that reducing the initial O vacancy density in MgO suppresses the formation of additional O vacancies and O interstitials, thereby mitigating resistance drift and TMR degradation. These pre-breakdown characteristics are consistent with the experimental results obtained under a constant voltage (Fig. 2(b)). With further calculations, we find that the breakdown lifetime strongly depends on the formation energy of oxygen defects.

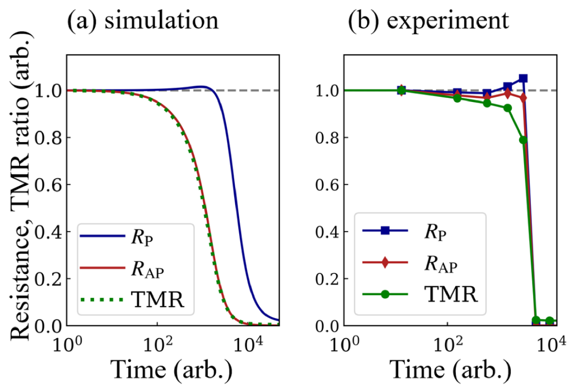

Fig. 2. Time dependence of the resistance in the P and AP state (R_P, R_{AP}) and TMR ratio under a stress voltage, obtained from (a) our breakdown model and (b) experiment. For the experimental data, the fabrication and measurement method of MTJs are reported in [8]. Adapted from [7]. ©2026 IEEE.

We investigate the impact of impurities on the breakdown lifetime, focusing on the changes in the O vacancy formation energy at sites neighboring an impurity atom as shown in Fig. 3(a). Fig. 3(b) shows the O vacancy formation energies for different impurity atoms, indicating that highly electronegative impurities, such as Ir and Pt, lower the O vacancy formation energy. These impurities reduce the net negative charge on the neighboring oxygen atoms, resulting in weaker ionic bonding and thereby lowering the formation energy of O vacancies. Fig. 3(c) shows the dependence of the lifetime on the impurity density obtained from our model. The lifetime decreases as the Ir density increases. Therefore, reducing the density of highly electronegative impurities in the MgO barrier is critical for improving breakdown lifetime.

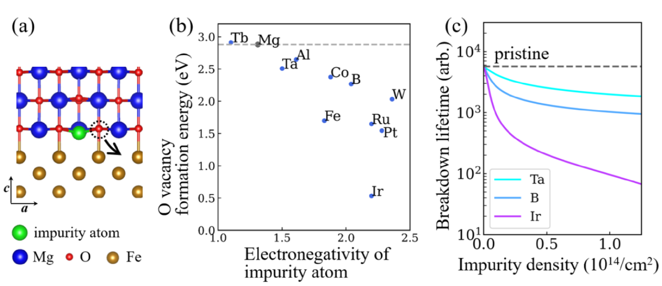

Fig. 3. (a) Oxygen site for evaluating vacancy formation. (b) Formation energies of O vacancies for different impurity atoms. (c) Dependence of simulated lifetime on the area density of impurities in MgO. Adapted from [7]. ©2026 IEEE.

IV. Conclusion

Using DFT calculations, we have developed a microscopic model for stress-induced degradation and breakdown of the MgO barrier. Impurities with high electronegativity, such as Ir and Pt, reduce the breakdown lifetime by lowering the formation energies of oxygen vacancies. These elements are commonly used in MTJs, and their incorporation should be carefully controlled during device fabrication. Using the same approach, we have also demonstrated that resistance drift and the time-dependent degradation of the TMR ratio can be attributed to the generation of oxygen defects at the MgO interface. Reducing the initial density of oxygen vacancies in MgO suppresses further vacancy formation and mitigates resistance drift and TMR degradation. These findings provide a fundamental basis for understanding the endurance of scaled MTJs and offer guidelines for optimizing STT-MRAM devices.

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